



FLEET & SHORE PERSONNEL TRAINING

Operation in Ice

Preparing the ship, the machinery and the crew for operation in ice and extreme cold weather.



IMO Polar Code

IMO Arctic Guidelines

SOLAS / MLC 2006

ISGOTT

Company SMS

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Operation in Ice

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02

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SECTION 01

Operation in Ice

Ice is potentially lethal to a ship that is not adequately prepared. Every operation in ice begins with a risk assessment.

01

Risk assessment before operation in ice

According to Company policy all ships shall assess risks prior to commencing operation in ice and cold weather.



Assess before you enter

Because of the natural risks and dangers of operation in ice and cold weather, a risk assessment is essential and is not optional.



Send the form ashore

The completed risk assessment form must be sent to the office before commencement of operation in ice.



Expect instructions back

Where additional instructions and assessments apply, office staff will send them back to the ship.

What cold-region risk management must cover

Risk management for cold regions should cover all hazards that may result from, or be accentuated by, the cold temperature environment and by operations in ice-infested waters.



Operability of every system

Verify the operability and check the functionality of all operational and safety systems, including firefighting and lifesaving.



Hull integrity

Monitor hull integrity by sounding tanks and voids at regular intervals.



Water temperatures

Monitor water temperatures in tanks and systems to provide warning of freeze-up problems.



Pollution prevention performance

Monitor the performance of pollution prevention systems to prevent the discharge of wastes.



Systems near their limits

Risk management requires an understanding that many systems and components will be operating at or near their design limits, and their performance may degrade rapidly. Monitoring and testing should be conducted more frequently than is the case in normal operations.

THE GOVERNING IDEA

Ice is potentially lethal to a ship that is not adequately prepared

The ice environment is so far removed from normal blue water shipping that adaptation by the crew to the conditions, as well as changes to the ship, are essential.

Prepare the ship

Prepare the machinery

Prepare the crew

Two things must change before entering ice

Preparation ranges from deck and engine work to hull integrity, navigation, accommodation and the experience of the crew.



Changes to the ship

Construction and equipment

- Ships should be suitably constructed and equipped for navigating through ice and operation in ice
- Engines, steering gear and navigation equipment are essential elements for carrying out the voyage
- Hull integrity must be verifiable throughout the passage
- Accommodation and deck areas must be protected against freezing



Adaptation by the crew

Experience and routine

- The crew must adapt to conditions that have no equivalent in blue water trade
- Navigation in Ice and Operation in Ice training is given before joining
- Watch routines, work periods and rest breaks all change in the cold
- Inexperienced officers require onboard familiarisation on unfamiliar equipment



SECTION 02

Ice Class Rules and Notations

Ship classification rules for vessels operating in ice exist to ensure that they can operate safely.

02

What all societies' ice class rules govern

New ice strengthening rules are being introduced, and new construction and designs are being developed to cope with increasing demands.

- Horizontal ice belt around the hull
- Machinery output, shafting and propeller
- Towing and mooring arrangements
- Provision of efficient ways to prevent freezing in ballast tanks located above the waterline
- Seachest suction arrangement
- Air capacity for starting compressors
- Rudder and steering gear
- Corrosion protection



Where to find your own figures

Notations are assigned by classification societies. The principal particulars belonging to each vessel can be found in her classification society documents, certificates, plans and manuals.

What makes a ship eligible for ice class

Eligibility is based on, but not restricted to, the following factors.



Reduced frame spacing

Closer framing to carry the ice loads on the shell plating.



De-icing equipment

Fitted to specific deck areas and to navigation equipment.



Increased power level

Enough power to maintain continuous progress in the ice conditions appropriate to the class.



Cold weather equipment

Special cold weather equipment carried for use by crew members.



Special materials

Materials such as special paint coatings that are not made brittle at cold temperatures.



Cooling intake design

Seawater cooling intakes provided with high and low suctions and internal circulation arrangements.

Two categories of ice notation

The notation of the ship is determined by the trade for which the vessel is intended. Notations fall into two categories, according to where additional strengthening is added.



First-year ice

Winter navigation only

Additional strengthening for navigation in waters with ice conditions during winter only. This covers the great majority of Baltic and North Atlantic winter trading.



Multi-year ice

Arctic and Antarctic

Additional strengthening for the multi-year ice found in Arctic or Antarctic regions, where the ice has survived at least one summer melt season.

Pollutant separation and double bottom rules

Requirement	Rule
Pollutant against the shell	No ice class ship should carry any pollutant directly against the outer shell.
Minimum double skin width	Any pollutant should be separated from the outer shell by double skin construction of at least 760 mm in width.
Double bottom extent	All ice class ships should have double bottoms over the breadth and the length between forepeak and afterpeak bulkheads.
Double bottom height	In accordance with the rules of the classification societies in force.
Working liquids exemption	Double bottoms should not carry pollutants unless a complying double skin is provided, or working liquids are carried in way of main machinery spaces in tanks not exceeding 20 m ³ individual volume.
Polar Classes 6 and 7	Double bottoms may be used for any working liquids where the tanks are aft of midships and within the flat of bottom.

Polar Class ships with icebreaking bow forms and short forepeaks may dispense with double bottoms up to the forepeak bulkhead in the area of the inclined stem, provided the watertight compartments between the forepeak bulkhead and the bulkhead at the stem-to-keel junction are not used to carry pollutants.



SECTION 03

Machinery and Systems Preparation

The engine must be reliable and responsive to bridge orders whenever needed. Everything else follows from that.

03

Engine preparation for operation in ice



Reliable and responsive

The engine must be reliable and responsive to bridge orders whenever needed. Manoeuvring demand in ice is far higher than on a conventional passage.



Sea suction filters

Filters to sea suctions must undergo regular inspections by watchkeepers, and any danger of overheating must be countered by effective watchkeeping duties.



Access to suction strainers

Access that allows easy removal and cleaning of suction strainers is useful for ships operating within the ice environment.



Purifiers and injection elements

Purifiers and fuel injection elements must be well maintained to ensure continuous and reliable operation of the main engines.



Below minus 30 °C

Main engines on ships operating at minus 30 °C or less are exposed to operational conditions and events that do not normally occur on conventional trade routes. Auxiliaries are exposed to the same conditions.



Shaft seals

Shaft seals are frequently damaged in ice operations and should be watched closely.

Steering, navigation and communication

All three systems are tested before entering ice limits, and again during navigation and operation in ice.



Steering gear

- Must be capable of quick and effective response
- Emergency and auxiliary steering systems tested before entering ice infested regions



Navigation equipment

- All instruments operationally tested before entering ice limits
- Tested again during navigation and operation in ice
- Any defects corrected where possible
- Inexperienced officers given onboard training on new or unfamiliar equipment



Communication

- Internal and external systems should be without defect
- Back-up systems tested periodically to ensure reliability

Fuel, water, provisions and searchlights

The Master is responsible for ensuring that there are ample reserves of all essentials on board. If a ship becomes fast in the ice, it may be some time before an icebreaker can reach it.



Twenty days of reserves

It is a Company requirement that whenever operating in ice, vessels keep a minimum of 20 days of reserves of fuel, water and provisions, to maintain life support and ship integrity in the event of prolonged entrapment by ice.



Searchlights

Sound searchlights are needed to allow the ship to continue at night, and all ice class ships must be fitted with them. Appropriate checks must be made to confirm that searchlights are working, in good condition and always ready for use during operation in ice.



SECTION 04

Ballast, Trim and Water Systems

Freezing ballast expands, and expansion cracks or distorts tank structure. The limits exist for that reason.

04

Freezing limits every officer must know

90%

Maximum ballast tank fill during freezing conditions, to allow for expansion

−1.8 °C

Temperature at which ice forms in seawater of density 1025 (28.8 °F)

−2 °C

Sea temperature at which the 90% limit and air bubbling apply

−30 °C

Lowest temperature at which bubble systems have been used satisfactorily



Air bubbling — At sea temperatures of −2 °C and sub-zero air temperatures, ballast tanks should not be filled more than 90% and the air bubbling system should be operated. The same applies when loading cargoes at a sub-zero temperature.



Planned maintenance — Vessels should carry out maintenance and routine checks of onboard systems as per PMS to ensure that they are in operating condition.

How and where ballast freezes



Where freezing starts

In tanks above the waterline, freezing starts at the top of the tank and at the side walls. Ice can also form on steelwork, sounding pipes, air pipes and valves.



Wind chill puts high tanks at risk

Because of the wind chill effect, ballast tanks above the waterline are at greater risk than tanks below it. Tanks positioned close to the ship's side are especially exposed.



Expansion damages structure

Freezing tanks can cause expansion and subsequent cracking or distortion of the tank structure. Valve and suction line freezing can occur even when the tank itself does not freeze.



Blocked air and vent pipes

In extremely cold regions, thick ice formation or complete blockage within air and vent pipes has been noted.



Salinity matters

Fresh or brackish water freezes more easily, so higher salinity sea water should be used where voyage routing and environmental regulations allow.



Ice is deadfreight

Any ice that forms in ballast tanks equates to a deadfreight value — weight that may not be dischargeable when the vessel is loading, reducing deadweight capacity.

Ballasting and trim in ice

✓ DO

- Ballast the ship as deeply as possible, allowing for freezing of unheated ballast tanks
- Circulate the ballast water to keep it moving and prevent freezing where tanks are unheated
- Provide some form of heating to ballast tanks above the waterline if prolonged navigation in ice is expected
- Keep the propeller as deep as possible, allowing for the thickness of ice that may be encountered
- Consider ballasting the holds when in ballast, if the ship's stability can safely allow it
- Expect the ice itself to act as an insulating layer — a sizeable tank rarely freezes solid

⊘ DON'T

- Do not fill ballast tanks to more than 90% during freezing conditions
- Do not assume a tank is safe because it has not frozen solid — valves and suction lines freeze first
- Do not carry fresh or brackish water ballast into a cold weather zone if sea water can be used instead
- Do not overlook falling ice chunks from tank sides after discharge of liquid ballast — they may damage coatings or components
- Do not trim the ship in a way that lifts the propeller tips towards the ice

Draining down before the cold



Fresh water tanks at 75%

Fresh water tanks such as those in the lifeboats, or drums where fitted, should be no more than 75% full. Fleet vessels normally carry fresh water in packs that have sufficient volume for expansion.



Drain exposed waterlines

All exposed waterlines, both salt and fresh water, should be drained down.



Drain the fire line completely

The fire line must be drained completely before arrival in cold weather areas and all exposed valves left cracked open. If this is not done, moisture within the valve can freeze and make it impossible to open.



Drain in under ten minutes

After use in cold conditions the fire line must be drained quickly — in less than 10 minutes.



Two valves stay shut

- Once the line is drained, all exposed valves must be left cracked open, except:
- The valve for washing anchors — avoids ice accumulation in the anchor hawse pipe and an accidental pressure drop if a fire occurs
- The isolating valves in the forecastle

Rapid draining of the fire line

To achieve rapid draining in less than ten minutes, open the following valves fully. The airflow from the upper valves accelerates the speed of drainage.



1. Lowest main deck valve

Open the lowest valve on the main deck fully so the line drains from its lowest point.



2. Upper bridge deck valves

Open the upper exposed bridge deck valves fully to admit air at the top of the system.



3. Upper forecastle valves

Open the upper exposed forecastle valves fully to complete the airflow path forward.



4. Crack the valves open

Once drained, leave all exposed valves cracked open — except the anchor wash valve and the forecastle isolating valves.



SECTION 05

Sea Inlets and Cooling Water

Blocked sea inlets are the most common ice problem of all — and they end in a dead-in-the-water scenario.

Frazil ice

Frazil is a type of ice that receives little attention even though it can cause great difficulties.



What it is

Suspended ice crystals, usually less than half a millimetre in length, found in the upper few metres of water exposed to sub-zero temperatures.



Chemically active

The crystals are very low in salinity but carry high concentrations of ions that alter the bulk electrical and chemical behaviour of the water on their surfaces.



Widespread

They are widespread because they are a transient stage in the formation of a sea ice crust, and matter most in areas of open water.



Rapid solidification

Under the right conditions frazil brings about rapid solidification. Agitation of suspended crystals in pumps and at intakes leads to rapid consolidation.

What happens when ice or slush enters the sea chests



Flow obstruction

If ice or slush enters the sea inlet chests, the flow of seawater to the engine cooling system can be obstructed. The majority of ships entering ice-covered waters meet this problem.



Overheating of prime movers

Restricted cooling water leads directly to overheating of the prime movers.



Loss of propulsion power

The result is a loss of propulsion power, creating a dead-in-the-water scenario in the middle of an ice field.



Draught helps

Deep draughted vessels are less affected by these conditions. All ice class fleet vessels normally have proper design features that prevent the build-up of ice in these circumstances.



Supervision is mandatory

Ships operating in ice must supervise the engine cooling system and prevent it from becoming blocked. Monitoring seawater pressure at the inlet is the best check for the presence of ice in the circuit.

Limiting ice ingress under all arrangements

Item	Setting or action
Sea strainer inlet water	Water coming into the sea strainer must be kept above 20 °C.
High sea suction	Keep the high sea suction closed.
Low sea suction	The low sea suction may need to be slowed.
Overboard discharge	Discharge overboard should be almost closed, to obtain positive recirculation.
Number of sea chests	It is advisable to use only one sea chest at a time.
Ice detection	Monitor seawater pressure at the inlet. Fluctuations and eventually a significant pressure drop indicate that it is time to clean the strainer.

Four ways to reduce seawater pumped into the cooling system

To avoid ice accumulation in the strainers it is important to reduce the quantity of seawater being pumped into the cooling system.



Throttle the discharge

Reduce the rate of flow through the seawater pump discharge valve.



Use a smaller pump

Use a lower capacity pump instead of the main seawater pump.



Reduce other coolers

Reduce the flow of cooling water to all other cooling systems in the engine room.



Steam injection

If the ship has no direct steam injection into the sea chests, a steam hose can be run. The use of steam alone is not enough to melt ice, even under light ice conditions.

Cleaning a seawater strainer safely

Cleaning the strainer is a risky operation and must be carried out with great caution after risk assessment.

1

Change over and isolate

Engine room personnel must use alternate sea chests and isolate the one about to be cleaned.

2

Prove the valves are tight

Take care that inlet and outlet valves are completely shut and tight before removing the strainer cover, as ice may remain on the valve seats. If so, repeat opening and closing of the valves until the ice clears out.

3

Flush out remaining ice

After opening the strainer, open the sea inlet valve slightly to flush out any remaining ice. A large quantity of ice and water can easily accumulate in engine room bilges.

4

Replace the cover

Replace the strainer cover after the operation. It may be necessary to repeat the operation a number of times.

5

Keep tools at each location

To reduce downtime at each strainer, each location should be equipped with the necessary tools and accessories.

Sea inlets, strainers and ballast tank vents

✓ DO

- Use only one sea chest at a time and keep the other available as a change-over
- Monitor seawater pressure at the inlet as the primary indicator of ice in the circuit
- Carry out a risk assessment before opening any strainer
- Remove the air pipe vent on ballast tanks to prevent the ball sticking
- Where removing the vent is not practical, a manhole may be opened instead
- Assign a person with a radio to stand by continuously at any open tank, in communication with the officer at the ballast console

⊘ DON'T

- Never use a seawater strainer without its perforated plate properly installed inside — the plate acts as a filter and prevents large chunks of ice entering the circulating system
- Do not rely on steam alone to melt ice, even under light ice conditions
- Do not open a strainer cover before proving both inlet and outlet valves shut and tight
- Do not leave an opened ballast tank unattended at any time



SECTION 06

Deck Protection and Equipment

Covers, grease and housekeeping decide whether deck machinery still works when it is needed.

06

What covers achieve, and what to make them from

Covers positioned over exposed deck equipment achieve two goals: they prevent water from getting on and into the equipment and freezing, and they allow the ice to be removed easily.



Use PVC or plastic

PVC or plastic is a better fabric to use than canvas for covering exposed deck equipment.



Avoid natural fibres

Avoid material made from natural fibres for painters, embarkation ladders, ropes or protective tarps. Natural fibres absorb moisture and lose flexibility and durability in cold weather.



Lag the steam pipes

Steam pipes on deck should be kept lagged, and the condition of the lagging should always be monitored for deterioration.

Equipment for which covers should be considered



Open boats

Open survival craft collect spray and freeze into their stowed position.



Winches and windlasses

Cover the drums and control gear so that ice can be lifted off rather than chipped off.



Pilot ladders

Pilot ladders should be covered and should not be lowered too early, as ice accumulates from spray.



Sounding and filling pipes

Cover sounding pipes and filling pipes so the caps and threads remain workable.



Ventilators and hatch accesses

Keep ventilators and hatch accesses covered so they can still be opened when required.



Machinery control areas

Cover machinery control areas and control boxes exposed on deck.

Scuppers, grease and hydraulics



Scuppers

- Clear the scuppers of ice and insert rubber hose piping into them
- When the pipes are frozen, the rubber tubing can be removed to reveal ice-free scuppers
- This enables water on the decks to drain away



Low temperature grease

- Use on exposed moving parts such as windlasses, winches, door hinges, davits and cranes
- Prevents water from entering the moving parts and freezing
- Should not be applied where oils are used, as this will clog the machinery
- Check maintenance manuals for advice before using these greases



Hydraulic tanks and lines

- Where there is no heating system for hydraulic tanks, fill with oil suitable for cold weather conditions
- Any exposed piping should be insulated
- Portable heaters are ineffective at keeping deck equipment ice-free
- They are often used in deck houses containing machinery, particularly on ships that do not regularly enter ice

Lashings, ropes and moving parts

✓ DO

- Stow all loose lashing material on deck away in bins and stowage racks, as far as practicable
- Grease all lashing material in use liberally — turnbuckles, shackles and similar
- Use polypropylene and other synthetic ropes, which are best suited to severe temperature use
- Keep all exposed movable parts liberally covered with grease: butterfly nuts and bolts, fire flap hinges, vents, valve spindles, sounding pipe covers, hydrant wheel spindles and steel door dogs
- Mix a little antifreeze into the grease — it is very effective
- Grease pilot door wheels and rams, and pilot ladder rollers and trackways

⊘ DON'T

- Do not use manila ropes for any lashings on deck — manila becomes stiff and impossible to handle
- Do not use natural fibre material for painters, embarkation ladders or protective tarps
- Do not apply low temperature grease where oils are used, as it will clog the machinery
- Do not leave lashing material loose on deck where it will freeze into the deck

Daily deck routine in the cold



Clear decks and walkways

Regularly clear decks and walkways between hatches of snow and ice.



Keep the tools to hand

Keep shovels, crow bars, hammers, spikes, sledge hammers, pickaxes (fire axes will suffice), a grease pot, a blow lamp and sufficient salt handy, in convenient sheltered locations near work areas.



Taking fresh water alongside

When receiving fresh water alongside, leave the water running continuously and ensure the end of the hose is pushed well down inside the fresh water tank filling pipe to prevent freezing.



Store room heating

Ensure heating, where fitted, is kept in operation in the storage rooms.



Strip the bilges dry

Hold bilges, store bilges, chain lockers and the bilges for the side passageways are to be stripped dry.



Protect sounding pipes

Use a heaving line, rubber hose or environmentally safe antifreeze in sounding pipes to prevent bursting of pipes.



SECTION 07

Ice Accretion and Stability

On a large ship it is easily possible to accumulate 300 tonnes of ice in a twenty-four hour period.

The stability figures for icing

For any ship operating in areas where ice accretion is likely to occur, adversely affecting stability, icing allowances should be included in the analysis of conditions of loading.

30

kg/m² on exposed weather decks and gangways

7.5

kg/m² for projected lateral area of each side of the vessel above the water plane

5%

Increase to projected area of continuous surfaces; 10% to the static moments of that area

300 t

Ice easily accumulated on a large ship in a 24-hour period



Discontinuous surfaces — The projected lateral area of rails, sundry booms, spars (except masts) and rigging of vessels having no sails, and of other small objects, is computed by increasing the total projected area of continuous surfaces by 5% and the static moments of that area by 10%.



Built into the design — These allowances are all considered during the initial design of ice class vessels. Where additional calculations are needed, they are carried out as described.

Full and half icing allowance

There are two levels of ice allowance, and which one applies depends on the geographic area of operation.



Full icing allowance

The severe case

- All exposed horizontal surfaces carry an ice weight of 30 kg/m^2
- All vertical surfaces carry a weight of 15 kg/m^2
- The position of the centre of gravity is calculated with regard to the heights of the ice-covered surfaces
- An additional percentage increase accounts for ice forming on cranes, derricks and rigging



Half icing allowance

The moderate case

- Applied in the geographic areas that do not attract the full allowance
- Vessels intended for operation where ice is known to occur are designed to minimise the accretion of ice
- They are equipped with means for removing ice as the Administration may require — pneumatic devices, or tools such as axes or wooden clubs for removing ice from bulwarks, rails and erections

Where full icing allowance must be made

Region	Definition
Eastern North Atlantic	The area north of latitude 65°30'N, between longitude 28°W and the west coast of Iceland, north of the north coast of Iceland.
Norwegian and Barents Seas	North of the rhumb line running from 66°N 15°W to 73°30'N 15°E; north of 73°30'N between 15°E and 35°E; and east of longitude 35°E.
Baltic Sea	North of latitude 56°N in the Baltic Sea.
Western North Atlantic	The area north of latitude 43°N, bounded in the west by the North American coast and in the east by the rhumb line from 43°N 48°W to 63°N 28°W, then along longitude 28°W. Includes Baffin Bay and Davis Strait.
North of North America	All sea areas north of the North American continent and west of the North Atlantic areas defined above.
North Pacific and Southern Ocean	During the icing season: the Sea of Okhotsk, the Gulf of Tartary and the Bering Sea. In addition, all areas in the southern hemisphere south of latitude 60°S.

What influences the amount of ice accretion



Wind force

The wind lifts spray and drives it over the ship.



Air temperature

Governs whether spray freezes on contact.



Ship's course

Course relative to wind and sea decides how much spray comes aboard.



Ship's speed

Higher speed drives more spray over the bow.



Wave height

Higher seas break more water over the ship.



Sea surface temperature

Colder water freezes sooner on contact with the structure.

When spray freezes on contact

Condition	Threshold
Open sea water	Seawater spray will freeze on contact with the ship when the air temperature is less than -6 °C and the sea temperature is less than -1 °C.
Frequency	During the winter months these values are regularly exceeded in the Pacific, Atlantic and Baltic areas.
Brackish and fresh water	In the Baltic and the Great Lakes, which have brackish to fresh water, icing will occur when the temperature is only just below 0 °C.
Warmer water	Icing can also take place at higher water temperatures if the air temperature is below 0 °C — the air cools the water breaking over the ship and ice is formed.
Where the ice builds	The bulk of the ice will be forward and on the windward side.
Reducing the rate	The accretion rate can be lessened by altering course, reducing speed, or a combination of both. In severe storms, even with the ship hove to, the rate of ice accretion will quickly increase.

What accreted ice does to the ship

**Stability is reduced**

As the layer of ice grows, a displacement in weight takes place. This displacement reduces the stability of the ship. Any reduction of water from a full ballast tank also presents a free surface effect.

**Trim by the head**

A build-up of ice on the forecastle will cause a change of trim by the head and the propeller to rise.

**Anchor equipment disrupted**

Accumulation in the fore end clogs hawse pipes and spurling pipes on the forward mooring deck and disrupts the anchor equipment.

**Scuppers blocked**

Deck scuppers may become blocked and fail to shed water from the fore end. Fully loaded ships lie low with limited freeboard and take considerable sea spray.

**Air pipes and P/V valves**

Tank systems fitted with air pipes and pressure vacuum valves may become covered in ice and cease to function effectively if adequate equipment is not used or proper mitigating measures are not taken.

**Doors and deck gear**

General deck equipment, including access doorways to pump rooms and store rooms, may be directly affected by freezing conditions, making their operation difficult.

THE STABILITY WARNING

Asymmetric icing can put the ship on a list and leave it vulnerable

Where only one side or part of the ship acquires a degree of icing, the ship can generate a list angle and may be left vulnerable to external forces such as wind and wave action.

Alter course

Reduce speed

Clear the ice early



SECTION 08

De-icing

Ice must be removed before it becomes a stability problem — but removing it is itself a hazardous task.

08

Salt and sand

A quantity of salt and sand, determined by the Master, will be kept on board. They are the most useful and easiest materials to use for de-icing.



Forward

Store salt and sand ready for use on the forward mooring deck, where ice accumulates first.



Aft

Store salt and sand ready for use aft, near the mooring station and access routes.



On the bridge

Keep a supply on the bridge for the bridge wings, where bare metal decks become dangerous.

Removing snow and ice from the ship



Shovels first

Shovels are essential for removing newly fallen snow before it freezes on top of the old. A path along the deck must be kept clear for personnel to move around.



Turn the ship for shelter

When it is necessary for the crew to remove ice accretion, the ship should be turned to provide as much protection from adverse weather as possible.



Wooden mallets and crowbars

Ice can be knocked off using wooden mallets, and suitable pickaxes and crowbars for tanker ships — but these must be used with care because of the damage that can be done to the fabric of the ship.



Brittle fracture

Extreme temperatures can cause brittle fractures in cast steel structures when they are struck by hammers.



Stow the equipment enclosed

De-icing equipment should be stowed in an enclosed space, such as the forecastle or aft store, so that it is readily available for use.



Cushion the lower decks

Ice falling from a mast stay or shrouds can easily penetrate thin steel plating. When clearing upper structures, position cushioning such as wooden cargo pallets or cushioned canvas sheeting to cover lower decks.

Twenty minutes at a time

When crew are employed on ice clearance, exposure is limited and the work is rotated.

20 min

Maximum continuous period any person should work on ice removal

Rotate

A strict route of work must be employed so teams are relieved on time



Proper PPE at all times — Proper PPE should be used at all times, as required by Company policy.



Watch above you — When working on deck in the ice, care must be taken to avoid ice falling accidentally from the rigging and masts. Much of this ice can be shaken off by hand or knocked off with mallets.



Clear before it consolidates — Remove newly fallen snow before it freezes on top of the old layer — it is far harder to remove afterwards.



SECTION 09

Safety in Ice

External safety equipment will be severely affected by ice — and it must still be ready for use.

09

Equipment, training and personnel

When dealing with safety in ice, the issues of equipment, training and personnel must be considered a priority.



Expect severe effects

External safety equipment will be severely affected by ice and, although this will cause considerable difficulties, it must be ready for use.



Periodic inspections continue

Periodic inspections of all safety-related systems will continue as required by the PMS during exposure to extreme temperatures, to confirm the effectiveness of the precautions being taken.



Additional inspections

If deemed necessary by the Master, additional inspections will be carried out.



Training before joining

Necessary training in Navigation in Ice and Operation in Ice is given to all related crew before they join the vessel, in line with Company training policies.

THE ABANDONMENT PRINCIPLE

The ship is the best lifeboat

When its integrity is unsustainable, the ship must be abandoned — but in the high latitudes the greatest threat to life is the cold, and the ship is likely to provide life support facilities well in excess of those provided by lifeboats or life rafts.

Stay with the ship if you can

Extreme cold is the killer

Abandon only when integrity fails

Preparing lifeboats and life rafts



Grease every moving part

All moving parts of the lifeboat and life raft davits must be greased, including all the cables. Enclosed lifeboat access door hinges and retaining brackets should also be greased.



Petroleum jelly on windows

Petroleum jelly should be applied to the plastiglass helmsman's view window and any other windows.



No scrapers or de-frosting sprays

Scrapers and de-frosting sprays should not be used on these windows — if they are plastic, this can cause damage.



Cover the safety equipment

All safety equipment must be covered, preferably with an artificial material rather than canvas, to limit moisture absorption.



Ice removal mallet nearby

Lifeboats and launching equipment must be kept free of ice accretion. An ice removal mallet should be available in the vicinity of lifeboats and life rafts, or electrical trace heating provided to protect critical items such as door seals.

Launching survival craft in high latitudes



Ice below the launch position

Launching into clear water may be problematic because of heavy ice floes concentrated at surface level below the launch positions.



Use the recovery davit

For this reason, and also to avoid damage, it is more prudent to launch free-fall lifeboats by lowering them with the recovery davit rather than by the free-fall method.



Free-fall trackways

Free-fall launching trackways need to be clear of ice formations to function correctly. Ice accretion is a major obstruction to launching devices and must be cleared well beforehand.

Using life rafts in ice

In ice, life rafts are not to be used unless there is no alternative.

✓ DO

- In heavy ice cover where boats cannot be launched, lower the life raft containers onto the ice, carry them to a safe ice floe and inflate them on the ice for shelter
- Where the ship is beset by ice, use lowered boats and rafts as shelter for those who have left the ship
- Where there is water or thin ice around the ship, boats may be used — but take care not to rub against any ice present
- Remember that the davit-launched survival raft keeps people dry throughout the evacuation procedure

⊘ DON'T

- Do not use life rafts in ice unless there is no alternative — ice can pierce the rubber
- Do not allow the abrasive ice surface to rub against the raft: friction weakens the fabric and eventually leads to a puncture
- Do not expect lowering the lifeboats to serve any purpose other than shelter when the ship is beset by ice
- Do not allow ice accretion to increase on or around survival craft

Starting and running in extreme cold



Start before taking to the water

It is normal practice to start lifeboat engines prior to the boat taking to the water. Boat engines are expected to start with fuel, lubricating oil and cooling fluid at very cold weather conditions.



Sub-zero fuel and oil

Fuel and engine oils used in lifeboats and rescue boats should be of a type suitable for operation in sub-zero temperatures. Data sheets should be available on board for verification.



Additives and heating

Where additives are used for this purpose, the Company will provide guidance in line with manufacturers' instructions. Where fitted, the heating system should be checked to be operational.



Stored energy devices

Particular attention should be given to avoiding loss of performance of batteries or other stored energy devices.



Propellers and brittle fracture

Propellers, although usually manufactured in bronze alloys, may exhibit brittle fractures in extreme cold. Ensure rotating propellers do not contact heavy ice floes. Some propellers have ducting that gives limited protection.

Speed and endurance in ice

6 kn

Maximum speed at which the boats can operate in ice conditions

24 h

Fuel is limited to operating at 6 knots for a 24-hour period



Master's judgement — Achievable speed will depend on the judgement of the Master, the surface ice type and its concentration.



Hypothermia in minutes — In cold conditions personnel can become hypothermic in minutes rather than hours. Self-help capability in extreme cold may be limited because of freezing muscles and limbs.



The real difficulties — The main difficulties during evacuation are the crew working in bulky winter clothing, on slippery decks, and the ice on railings and other superstructure areas that were not greased or protected.

Drinking water in survival craft



Tinned units in life rafts

Life rafts tend to have tinned fresh water units. The packs of water have enough space for expansion, but should be checked regularly in line with Company policy to ensure no packing is damaged by expansion.



Bulk drums crack

Drinking water contained in lifeboats in bulk drums is known to freeze at extreme low temperatures and can crack with the expansion of the frozen water.



Drain one third

To avoid damaged drums, the safety officer should normally drain a third of the water if the ship experiences freezing conditions. This reduces the water available in an abandonment situation, but the alternative would be no water at all.



Old ice or snow

Where additional water is required by personnel in lifeboats, old ice or snow, which contains less salt than new, should be used as a water source.



Move to a heated space

In case of need, lifeboat water containers should be moved to an adjacent heated space.

Rope ladders and thermal protective aids



Why the rungs are wooden

Rope ladders with wooden rungs are common to most lifeboats and provide a way of boarding for anyone in the water. They are usually fitted with gunwale hooks. The rungs are wood for two reasons: they float outwards towards anyone in the water, and wood would not freeze a survivor's hands as metal might in cold climates.



The non-ferrous last rung

The last rung is often made in a non-ferrous metal that causes it to lie below the surface, weighting the end of the ladder down slightly and making it easier for a casualty to climb up.



Thermal Protective Aids

TPAs should be verified as effective within a temperature range appropriate to the temperatures likely to be encountered.



Verify before you sail

Immersion suits and TPAs should be addressed, and equipment suitable for anticipated temperatures provided, prior to the vessel entering cold areas.

Why cold makes firefighting harder

Significant risks are associated with the use of firefighting equipment in extreme low temperatures, the most significant being the potential freezing of fluids in lines.



Frozen mains and nozzles

Freezing of fire water hoses, piping and nozzles. Fire mains are normally charged and pressure is maintained with a topping-off pump. At -30 °C this may have to be changed and the fire mains drained until needed.



Extinguisher storage frozen

Portable fire extinguisher storage may be obstructed or frozen shut.



Fire dampers frozen

Fire dampers may freeze in the stowage position.



Fire risk is higher

Fire is a considerable danger, heightened in the accommodation areas by the extra heaters and the load put on the electrical fuse boards and wiring.



Reliance on extinguishers

With the fire main drained down, reliance is on extinguishers for the initial firefighting, so prevention is essential. Firefighting outside the accommodation is even more of a problem when the main is drained.

Timings and settings for the fire main in sub-zero conditions

Item	Requirement
Fire patrol frequency	Fire patrol will be carried out at least every hour whenever operating in ice.
Draining after use	When the fire main is brought into use for a fire or exercise, it must be drained again within 10 minutes of completion.
Drained condition check	In sub-zero temperatures, drain the fire main from the lowest point and close the drain valve to keep the system ready. Check the drained condition by operating the drain valve every 1 hour.
Re-pressurising	The procedure should also consider the method of re-pressurising the fire main in sub-zero temperatures, with the aim of avoiding freezing of hydrants and monitors.
Change-over temperature	At -30 °C the topping-off arrangement may have to be changed and the fire mains drained until needed.
Night inspections	Further inspections at night are usually not feasible in the prevailing weather conditions found in ice waters.

Firefighting equipment in the cold

✓ DO

- Locate fire main isolating valves so that they are accessible, and protect any isolating valves in exposed locations from freezing spray
- Position or design hydrants to remain operable under all anticipated conditions, taking ice accumulation into account
- Use hydrant valves of a design that ensures ease of operation by personnel wearing heavy gloves
- Keep fire pumps, including the emergency fire pump, in heated spaces or otherwise protected from freezing
- Take account of the prevention of clogging of sea inlets with ice in the emergency fire pump arrangements
- Place any heaters used securely so they cannot move, in an area free of materials that could cause fire

⊘ DON'T

- Do not locate water or foam extinguishers in any position exposed to freezing temperatures
- Do not use electric bar heaters in the accommodation, owing to the fire risk
- Do not stow materials against bulkheads where heaters are in use
- Do not leave the fire main charged and unattended in sub-zero temperatures — drain it and check the drain valve hourly
- Do not assume fire extinguishing systems remain accessible: they must be designed or located so they are not made inoperable by ice, snow or low temperatures

Working on deck in extreme cold

Even normal work is severely affected by ice.



Salt and sand on walkways

Unless walkways are fitted with heating arrangements, salt and sand is the only way to keep the walkways used for essential crew movement clear and non-slip.



Bridge wings

Where the bridge wings have bare metal decks, wooden duck boards should be placed down, and a supply of sand and salt kept on the bridge for use.



Working over the side

Should it be necessary to have a crew member working over or near the side, a lifeline must be used. If he falls into the sea the chances of recovery are remote, and the extreme cold of the water will limit chances of survival.



Unnoticed lacerations

With extreme cold, an accidental laceration can occur without the person realising until blood is seen — or, as bleeding also slows in cold weather, until the person enters the warmth of the accommodation and removes protective clothing.



Cold metal burns skin

When metal objects are very cold they can cause severe skin burns, and must never be allowed to come into contact with the skin.

Accommodation, walkways and entrapment



Accommodation design

All personnel accommodation should be designed and arranged to protect the occupants from unfavourable environmental conditions and minimise risk of injury during normal operations — including ice transiting and icebreaking — and in emergency conditions.



Public spaces and equipment

Accommodation, public spaces and the equipment installed in them should be designed so that each person making proper use of them will not suffer injury during normal open water operations, designed ice transiting modes and emergency manoeuvring conditions.



Polar Classes 1 to 5

Ships of Polar Classes 1 to 5 inclusive should have sufficiently available and reliable facilities to maintain a life sustaining environment in the event of an emergency and/or extended ice entrapment.



Clearance equipment

Walkways and access ladders must be maintained in a safe condition. Vessels should be equipped with materials and equipment to clear snow and ice from critical areas — for example wooden mallets, heated stairs and handrails.



SECTION 10

Checklist for the Deck Department

Items to be checked prior to entering ice or carrying out operations in ice.

10

Deck department general checks

- Check the inventory of cold weather gear on board and replenish any deficiencies
- Where the ship is not fitted with double glazing, perspex or cling film over windows can help insulate the accommodation
- Examine towing fittings on the forecastle and ensure that a suitable messenger is available to pick up a bridle or towline
- Ensure regular monitoring of the water level in all spaces and compartments — as frequently as hourly in areas of heavy ice concentration
- Repeat that monitoring after any strong ice impact
- Searchlights fitted and tested
- Ship's hydraulic systems filled with a fluid suitable for cold weather operations
- Inspect all cargo, bunker, ballast and subsidiary valves required for operations before entering cold areas, to confirm their gearboxes contain no water and are well greased



Prove the anchors

It is recommended that both anchors are lowered to prove that they are free to run from the pipe, when safe navigation permits.

Thrusters, ladders and mooring lines

✓ DO

- Protect mooring wires and synthetic ropes to stop ice accretion until they are required for use
- Remember that ice crystals can form within unprotected ropes and damage the rope's fibres
- Cover all mooring lines on drums, leaving lashings clear before making fast to enable release
- Lower both anchors to prove they run free when safe navigation permits
- Keep the towing fittings on the forecastle examined and a messenger ready

⊘ DON'T

- Do not use tunnel thrusters — they are liable to damage from ingested ice, resulting in damaged blades
- Do not lower pilot ladders too early, as ice can accumulate from spray
- Do not leave valve gearboxes unchecked for water content before entering cold areas
- Do not leave mooring wires and synthetic ropes exposed on deck without protection

Lines to isolate and drain, leaving the drains open

Line	Action
Fire mains	Isolate and drain. Valves must be left cracked open.
General service and fresh water	Isolate and drain the general service line and fresh water line to all deck outside taps.
Deck air line	Close the main air valve to deck and drain the airline, taking care to remove any moisture in the line, especially at the ends. If air must be supplied to deck, an air drier may be used.
Chain locker eductor	Isolate and drain, leaving the drain open.
Anchor wash and winch cooling	Isolate and drain the anchor wash lines and the winch cooling lines.
External taps and window wash	Drain all external taps, and drain bridge window wash systems unless antifreeze has been added.

Control of drain valves will be made every 4 hours to avoid ice formation. Be alert for the formation of ice at any overboard discharges.

Two short checklists

Both lists are completed before the vessel enters cold areas.



Whistles and fog horns

Drain the main and forward whistles and close the drain cocks. Confirm that the air supply to the whistle is moisture free — the deck air line is drained at the engine room valve.



Life raft and LSA gear

Stow the forward life raft and LSA gear so it is not left exposed to freezing spray on the forecastle.



Extinguishers and lifebuoys

Stow deck foam and water fire extinguishers in heated areas, because water and foam extinguishers must never be left in a position exposed to freezing temperatures. Stow lifebuoys.



TPAs and immersion suits

Thermal Protective Aids and immersion suits should be addressed, and equipment suitable for anticipated temperatures provided, prior to the vessel entering cold areas.

Lifeboat checks and cold weather additions

- Drain fresh water tanks where applicable
- Ensure portable fresh water packs are not damaged; where drums are carried, ensure they are full and located in a warm room near the lifeboat
- Fit the drain plugs
- Consider adding sets of sub-zero temperature insulated clothing, including jackets with hood, trousers, gloves and boots that can be worn beneath immersion suits
- Consider adding sets of thermal underwear and socks
- Consider adding sets of face masks and goggles
- Consider adding down-filled sleeping bags
- Consider adding Arctic type igloo shaped survival tents
- Consider adding additional high energy food supplies, such as chocolate bars



Ice delays release

Lifeboats and life rafts covered in ice and snow will take much longer to release than if the build-up is removed regularly. Throughout the ship's time in ice conditions, lifeboats, rescue craft, life rafts and launching equipment must be regularly cleared — using wooden mallets stored close by, and without damaging the equipment.

Where grease must be applied liberally

Grease should be applied liberally to all moving parts on the decks. A little antifreeze mixed into the grease is effective.



Windlasses

All moving parts and the brake gear.



Mooring winches

Drums, warping ends and controls.



Cargo winches

Including the hatch drive gear.



Engaging clutches

And all pins and operating handles.



Brake clamping bolt threads

So the brakes can still be set and released.



All nipple points

Work every grease nipple on deck.



Sounding pipe cap threads

And hatch dogs.



Davits and cranes

Including all cables and sheaves.

Covers on, heaters running



Protection

Covers and lagging

- Covers off all searchlights
- Covers on hatch control boxes
- Covers on all fire hydrants
- Covers on all fire hose boxes
- Covers on all mooring lines on drums, leaving lashings clear before making fast to enable release
- Covers on bunkering sounding pipes
- Grease and burlap (hessian) hatch drive motors and gypsies



Heaters

Switch on before arrival

- Forecastle fan and space heaters
- Winch hydraulic header tank and hatch hydraulic heaters
- Anti-condensation heaters on hydraulic systems
- Winch, hatch pump and exhaust fan motor heaters
- Any deck house space heaters
- Fire pump and davit heaters
- Space heaters in emergency headquarters



SECTION 11

Checklist for the Engine Department

A continuous engine room watch begins at the ice edge and does not end until the ship is clear.

11

The engine room is never left unattended in ice-covered waters

A continuous watch in the engine room must be commenced when the vessel arrives at the ice edge. Engineers must be informed about the anticipated conditions according to ice charts and reports.

Watch from the ice edge

Brief on the ice charts

No unmanned periods

Engine department numeric requirements

Item	Requirement
Fuel and fresh water reserves	Ensure there is sufficient fuel and fresh water for at least 10 days.
Diesel oil settling and service tanks	Heat up to 30 °C.
Air pipes where no heating fitted	Where no heating arrangement is provided, the air pipes to settling and service tanks required for the main propulsion plant and essential auxiliaries are checked every hour to ensure they remain clear.
Diesel oil filters	Clean daily to prevent wax accumulation.
Purifiers	Lube oil and heavy oil purifiers should be run continuously.
Seawater temperature at inverter	Maintain seawater temperature at the inverter at 45 °F (7.2 °C).

Engine department general checklist

- Establish whether trace heating lines can be utilised
- Confirm whether combustion air preheating systems are operational for diesel engine jacket water heaters and other systems as applicable
- Compartment space heaters switched on
- Change over to the ballast seawater recirculating tank, if fitted
- Oil tank heaters on — keep the steam valve cracked open to prevent bursting of the pipe
- Keep all steam tracing lines active
- Keep the stern tube cooling water tank slack (after peak)
- Oily-water separator to be drained of all water if exposed
- Check the stern glands to determine whether water is passing through



Continuous watch

The engine room should not be left unattended in ice-covered waters. Continuous watch in the engine room must be commenced when the vessel arrives at the ice edge.

Keeping domestic systems flowing



Watch the pumps

Monitor domestic fresh water pumps — continuous running is an indication of ruptured piping.



Watch the tank levels

Watch the fresh water tank levels and crack the steam on the fresh water tank coils.



Circulate continuously

Circulate fresh water continuously to eliminate any possibility of freezing, and maintain a small flow of water through the sanitary system and soil drains.



Below 5 °C

If the ambient space temperature falls below 5 °C (41 °F), apply heating to the sanitary holding tank and the overboard line.



Purge the outlets

If time permits, purge air or steam through all wash water outlets that do not drain to a level well below the waterline.



Traps and toilets

If necessary, add a small amount of environmentally safe antifreeze or salt to unused drain traps and toilets, or insert rubber hoses for easy future removal.

Managing engine room ventilation in the cold



Shield the instruments

Reduce ventilation and/or provide shields to avoid any direct cold airflow against control and gauging equipment and small diameter piping systems.



Avoid recirculation

Recirculation of air in the engine room should be avoided, as this results in a build-up of aromatic hydrocarbons and noxious gases.



Funnel flaps

Close funnel flaps as required. Stop engine room vent fans and/or close distribution flaps as conditions dictate.



Funnel circulation

Leave the funnel skylights open to maintain funnel circulation, ensuring the funnel temperature is kept above freezing.



Fan heaters

Use engine room fan heaters as required.



Auxiliary boiler

Run on back-up if steam demand makes it necessary.

Recirculation and steam injection



Maintain 7.2 °C at the inverter

Maintain seawater temperature at the inverter at 45 °F, that is 7.2 °C.



Activate recirculation

Activate the seawater recirculation system by recirculating to the low sea chest and/or pump suction, as required.



Monitor pump pressures

Monitor suction and discharge pressures of the seawater pumps. Pressure and flow fluctuations indicate ice formation in the sea chest.



Steam injection ready

Carry out steam injection if any signs of ice formation are noticed in the sea chest. Steam injection hoses are to be kept ready. Open steam to the sea chests and the overboard discharge valve if required.



Sea inlets

Reduce the quantity of seawater pumped into the cooling system to reduce blockage by ice, monitor the pressure of seawater at the inlet, and clean the strainer as necessary.

Hydraulic pump rooms and steering gear flat

The same four measures apply to both compartments.



Hydraulic pump rooms

- Ship's hydraulic systems filled with fluid intended for cold weather operation
- Space heater switched on
- Oil tank heater switched on
- Compartment vent closed, or a shield must be used
- 70% isopropyl alcohol used to remove moisture in hydraulic oil and assist movement of the controls



Steering gear flat

- Ship's hydraulic systems filled with fluid intended for cold weather operation
- Space heater switched on
- Oil tank heater switched on
- Compartment vent closed, or a shield must be used
- 70% isopropyl alcohol used to remove moisture in hydraulic oil and assist movement of the controls



Passages and thruster room

- Under-deck passages and duct keel space: heater switched on
- Under-deck passages and duct keel space: compartment vent closed or a shield used
- Bow thruster room: space heater switched on
- Bow thruster room: oil tank heater switched on
- Bow thruster room: close compartment vent or use a shield

Emergency generator room and double bottom tanks

Space	Actions
Emergency generator room — fuel	Use winter grade oil.
Emergency generator room — antifreeze	Add antifreeze suitable to minus 40 °C to the generator and compressor.
Emergency generator room — vents	Close the external vents. Any fresh water lines to be blanked off. Radiator vents must be open for recirculation. Space heater switched on.
Double bottom tanks — fuel	Transfer maximum heavy fuel oil up to the wing tanks.
Double bottom tanks — heating coils	Isolate unused heating coils and blow through with air, or leave steam on and ensure returns are maintained warm.
Double bottom tanks — bilge tank	Turn on the bilge tank heating coil, unless the tank is empty.

Fire main, deck air and sea inlets



Fire main line

Put the pump suction on a low sea chest. Drain the fire main down to sea level. Leave the fire pump suction and discharge valves open.



Deck air

Isolate deck service air at the engine room valve and drain the line. Ensure that air supply to control air and the whistle is moisture free.



Sea inlets

Reduce the quantity of seawater pumped into the cooling system to reduce blockage by ice. Monitor the pressure of seawater at the inlet and clean the strainer as necessary.



SECTION 12

Batteries and Electrolytes

Battery performance is severely degraded in the cold, so battery handling becomes far more common — and far more hazardous.

12

Rechargeable batteries and their electrolytes

Because battery performance is severely degraded at cold temperatures, personnel using battery-powered equipment are more likely to handle batteries during a cold weather voyage than under normal conditions — recharging, warming up batteries that have become too cold, or replacing batteries that are no longer serviceable.



Lead-acid

The most common type aboard

Contains an electrolyte solution of sulphuric acid and distilled water. Sulphuric acid is very dangerous to personnel and highly corrosive to equipment or materials.



Alkaline

Nickel-cadmium

Contains nickel-cadmium and an electrolyte solution of potassium hydroxide. This electrolyte is also dangerous to personnel and is corrosive to many materials including aluminium and glass.

Sparks, flames and compartment ventilation

Both hydrogen and oxygen gas are given off from a storage battery, especially during recharging. Hydrogen mixed with oxygen or air is very explosive.

✓ DO

- Ventilate a sealed battery compartment thoroughly before it is entered — hydrogen is colourless, odourless and tasteless
- Verify that the ventilating system is operating properly before starting a charge
- Stop the charge if the ventilation is interrupted
- Ensure no current is flowing into or out of the battery before disconnecting or connecting terminals
- Where one terminal is grounded, disconnect the grounded terminal first when removing and connect it last when replacing
- Verify all terminal connections are tight to preclude sparks due to loose connections, and use only tools with insulated handles

⊘ DON'T

- Never allow sparks, smoking or flames of any kind in the vicinity of any rechargeable battery
- Do not turn on light switches in an unventilated battery compartment
- Do not make or break any electrical connection in the compartment until it has been well ventilated
- Do not charge a battery at a rate other than that given on its nameplate
- Do not use uninsulated tools near battery terminals — they can short-circuit the terminals

Charging and battery damage limits

Condition	Limit or action
Charging rate	Charge a battery only at the rates given on its nameplate.
Gassing or 35 °C	Reduce the charging rate if the electrolyte begins to evolve bubbles of hydrogen and oxygen, or if the battery temperature reaches 35 °C.
52 °C	Stop charging if the battery temperature approaches the upper limit of 52 °C.
Battery heaters	Equipment for batteries exposed to cold weather must be controlled so that the battery compartment stays below 35 °C and the battery below 52 °C, to prevent damage to the battery.
Discharged battery freezing	The sulphuric acid electrolyte in a discharged battery can freeze at –15 °C, whereas in a fully charged battery it remains liquid down to –51 °C or below.
Specific gravity	Sulphuric acid with a specific gravity greater than 1.350 must not be added to the battery. Only premixed electrolyte is to be used or stored aboard ship.

Freezing may damage the battery internally and/or crack the case, causing a hazardous spillage of electrolyte. All batteries exposed to cold weather conditions should be kept fully charged.

Handling electrolyte and preventing battery damage



Full protection

Personnel handling any electrolyte, or cleaning up a spill, should wear rubber aprons, rubber boots, rubber gloves and goggles or a full face shield, so that the electrolyte cannot come into contact with clothing or skin.



Adding water

When distilled water is added to a battery to replace that lost during normal operation, add it just before the battery is placed on charge. The water remains on top of the electrolyte until mixed by charging; in cold weather the unmixed water may freeze, cracking the case and leaking electrolyte.



Replacing spilled electrolyte

Where spilled electrolyte must be replaced, no distilled water should be added to the battery. Use only premixed electrolyte.



Concentrated sulphuric acid

The use and storage of concentrated acid for preparing electrolyte or adjusting specific gravity is authorised only for shore activities or ships designated as IMAs.

Treatment of electrolyte burns

1

Skin contact — flush 15 minutes

Should acid or electrolyte contact the skin, immediately wash the affected area freely with a large quantity of fresh water for about 15 minutes.

2

Eye contact — flush 15 minutes

Should acid or electrolyte contact the eyes, flush with fresh water for a minimum of 15 minutes, ensuring that both upper and lower lids are pulled sufficiently to allow water to flush under them.

3

Notify the medical department

In either case the medical department must be notified as soon as possible and requested to come to the site. If it cannot be notified, wash or flush with water for 15 minutes before transporting the victim to the sickbay.

4

Last resort water

In an extreme emergency, where fresh water is not available, seawater may be used — but only as a last resort.

5

Remove contaminated clothing

Clothing splattered with electrolyte should be promptly removed, and skin areas touched by electrolyte beneath contaminated clothing promptly flushed with water as described above.



SECTION 13

Crew Preparation for Ice

Clothes by themselves do not warm the body — the body warms itself, and clothing keeps that heat in.

13

Preparing the people before the passage



Training before joining

Necessary training in Navigation in Ice and Operation in Ice will be given to all related crew before they join the vessel, in line with Company training policies.



How clothing works

Clothes by themselves do not warm the body. The body is warmed by its own heat generation, with body heat warming the layers of air trapped between the skin and the clothing.



Company-supplied PPE

It is essential that crew wear protective clothing suitable for low temperatures, and the Company will supply proper clothing for all crew.

Duties that need special protection and managed routines

Particular attention should be given to ensuring that personnel with duties in exposed locations are suitably protected, through the provision of cold weather PPE and the management of work routines.



Bow lookout

The most exposed station on the ship, directly in the spray and the wind.



Gangway watch

Long static exposure in port, where the body generates little heat.



Bridge wing lookout

Exposed and static, often on bare metal decks unless duck boards are fitted.



Reference guidance

For guidance on PPE and working in exposed locations, refer to OGP Report 398.



Polar Service Temperature

Refer to the Polar Service Temperature — Polar Ship Certificate 2.3.1 and Polar Code part I-A section 1.2.11.

What makes each ice incident different

All ships will carry the emergency checklist, and the actions of crew members immediately following an incident will be in accordance with its content. The immediate actions of the Officer of the Watch, the Master and the Chief Officer must be practical and aimed at countering the greatest threat.



The weather

Prevailing conditions at the time of the incident.



Position of impact

Where on the hull the contact was taken.



The ship's speed

Speed at the moment of impact.



Its location

How remote the position is at the time.



Type of ship

Design, strengthening and cargo carried.

Why an ice incident is not a warm-water incident



It is likely to be very cold

Many actions are common from one situation to another, but in an ice environment a collision is likely to happen in extreme cold, and the survival techniques will be different to an incident in warmer waters.



Staying with the ship

The use of survival craft may not be a welcome option. Staying with the ship for longer prior to abandonment, provided circumstances allow, may be the better prospect.



The ship has better life support

The ship is likely to provide improved life support facilities well in excess of those provided by lifeboats or life rafts.



Damage is not the only risk

Accidents in ice conditions do not necessarily involve physical damage by the ice. Cold conditions combined with poor shipkeeping and inappropriate decision making make self-help and rescue more difficult.

What to wear, and what never to take off



Keep wet clothes on

If a person is exposed to the elements and there is no dry clothing to change into, wet clothes must be kept on. The insulating effect of wet clothing is greatly reduced compared with dry, but it is still considerably better than bare skin.



More layers, more insulation

More layers of clothing covering the body leads to more layers of insulating air to keep the body warm.



Never discard gloves or mittens

Gloves or mittens must not be discarded, whether in a lifeboat or in the water. Hands must be protected against the chilling effect of cold water so that people in the water can help themselves, for example by holding onto a lifeline.



On the general alarm

One of the first things to do on hearing the general emergency signal is to put on suitable clothing.



Immersion suits

Immersion suits are manufactured as a cover-all and include insulation for the head; the only exposed skin is the face. They are fitted with hand mittens on lanyards or pocketed gloves, and variations come with built-in waterproof boots or socks, a built-in lifejacket, a whistle and a light.

The metabolic cost of working in the cold

The added bulk and weight of cold weather clothing increases the metabolic requirements for any task.

25–50%

Increase in necessary calorie intake, depending on working conditions

36.8 °C

Normal body temperature — below this level hypothermia begins to set in
(98.2 °F)



Frequent breaks and hot food — It is important that frequent breaks are given, that crew and officers can enter the warmth of the accommodation, and that there is always an adequate supply of warm food and drink.



Hydration — It is particularly important to drink enough fluids, since workers may not realise they have become dehydrated.



Redundant heating — Accommodation heating systems should have adequate redundancy. External bulkheads within the accommodation should be insulated to prevent condensation in sub-zero temperatures.

Heating requirements for seafarer accommodation

- The system of heating the seafarer accommodation should be in operation at all times when seafarers are living or working on board and conditions require its use
- In all ships in which a heating system is required, the heating should be by means of hot water, warm air, electricity, steam or equivalent
- Within the accommodation area, steam should not be used as a medium for heat transmission
- The heating system should be capable of maintaining the temperature in seafarer accommodation at a satisfactory level under normal conditions of weather and climate likely to be met in the trade in which the ship is engaged
- The competent authority should prescribe the standard to be provided
- Radiators and other heating apparatus should be placed and, where necessary, shielded so as to avoid risk of fire or danger or discomfort to the occupants



Steam is the primary system

Steam heating to the accommodation is to be on. Steam is to be the primary heating system, backed up by individual cabin heaters where fitted. This helps to cocoon the accommodation spaces from the cold and prevents fresh water pipes from freezing.

Heating, moisture and access in the cold

✓ DO

- Cover portholes and window cavities with polyethylene or perspex to provide air buffer insulation where not double glazed or heated — this reduces ice formation on interior surfaces
- Keep all double doors to the accommodation, doors to stores, passageways and holds shut at all times
- Keep accommodation spaces in a state of light pressurisation and recirculate air
- Keep all steps, decks, walkways and passages into and around the accommodation clear of accumulated snow and ice by regular shovelling and salting
- Disperse coarse salt liberally on deck walkways, gangways and ladder steps to prevent slipping accidents
- Keep a sufficient stock of de-icing salt on board and readily available
- Place fan heaters by each accommodation door where extra heating is required

⊘ DON'T

- Do not use electric bar heaters, owing to the fire risk
- Do not introduce moisture into the air in cold climates where the ship is not insulated and provided with a vapour shield — vapour will condense between walls and ceilings, causing a fire hazard around electrical circuits
- Do not use steam as a medium for heat transmission within the accommodation area
- Do not leave the accommodation heating off at any time when seafarers are living or working on board and conditions require it

What cold does to the ability to work

Cold conditions severely hamper both physical and cognitive work. All of these effects should be considered during work planning.



Loss of dexterity

The decrease in dexterity, due to bulky clothing as well as the cold, makes it more difficult to carry out work needing fine motor skills, coordination and any sense of touch.



Musculoskeletal injury

Working in cold environments is related to an increase in the risk of musculoskeletal injury.



Cognitive decline lasts longer

The cognitive abilities of crew members will be affected long after a decline in dexterity or physical coordination.



Decision-making degraded

The decision-making skills of perception, memory and reasoning are all lowered in cold conditions, and tasks needing vigilance are made more difficult.



Heat loss without freezing

Even in conditions that are not freezing, wind and humidity take heat away from the body and this can eventually lead to hypothermia.



Protect the extremities

Cold initially affects the extremities that have less blood flow — hands, feet and head. These areas must always be protected.



SECTION 14

Immersion and Cold Injury

The body loses heat twenty-six times faster when wet than when dry.

14

What happens to the body in cold water

1

Initial immersion — cold shock

The sudden change to body temperature can cause severe stress to the heart, generating rapid uncontrollable breathing known as hyperventilation. Heart rate and blood pressure increase, with possible incapacitation or fatality due to cardiac arrest.

2

Short-term immersion — up to 15 minutes

A lifejacket is a considerable help, particularly if fitted with a splash guard. Survival time is greatly enhanced by not swimming, which conserves energy. Once the body has acclimatised, heat loss can be controlled by a reduction in activity.

3

Long-term immersion — 30 minutes or more

Core body temperature starts to reduce, to a greater or lesser degree depending on sea temperature and how well the individual is protected by layers of clothing. Where an immersion suit is worn, the rate of cooling is slower.

4

Post immersion

Many victims have died following recovery because of post immersion collapse. While immersed the body experiences a hydrostatic squeeze from the surrounding water pressure; after recovery that pressure is gone and the casualty can suffer cardiac arrest.

Why movement in the water is dangerous



Movement increases heat loss

Any movement in the water increases blood flow to the muscles and ultimately causes conductive heat loss from the exposed extremities. The surrounding water is stirred and convective heat loss from the body to the water takes place.



Conditions matter

The prevailing conditions at the time of immersion make a considerable difference. Remaining stationary might not be easy in rough sea conditions.



The first few minutes

The first few minutes of inactivity may allow the body to become acclimatised to the sudden change in temperature.



Hold onto something

Being able to hold onto a floating object and reduce movement may well reduce the possibility of hyperventilation.



Improving your chances

- Delay the time of immersion to the very last moment
- Don layers of clothing with a waterproof cover-all before entering the water
- Be educated and knowledgeable in self-help techniques

The numbers that define survival

35 °C

Hypothermia occurs at this body temperature; the average person loses consciousness

28 °C

Hypothermic cardiac arrest would not normally be expected before this temperature

26×

The body loses heat twenty-six times faster when wet than when dry

1 nm

Approximate detectable range of an approved lifejacket light



Below 28 °C — Once this body temperature is reached, death is almost inevitable.



Estimating survival time — Estimating survival time is difficult because of the many variables — for example the difference between a casualty who has received survival training and one who has not, or whether a lifejacket is fitted with a splash guard.



Stay still, stay calm — When all circumstances are taken into account, the person who survives longer is the person who remains as stationary and as calm as possible while immersed.

Why every detail on the lifejacket is there

Wearing a lifejacket is essential. An inherently buoyant lifejacket is better than an inflated one, as it is less likely to be affected by a puncture and is a better aid to warmth.



Retro-reflective tape

Fitted to all approved lifejackets to reflect a searchlight at night. It is also fitted to life rafts for the same purpose.



A plastic whistle

The whistle is manufactured in plastic, not metal, to avoid the lips freezing to the surface in cold climates.



A light

The detectable range of the light is about 1 nautical mile.



A towing loop

Found at the top of the jacket, to allow the wearer to be pulled through the water. It is not designed for lifting individuals clear of the water surface.

What to do — and not do — in cold water

Every person can improve their survival chances in cold or iced waters with knowledge and simple self-help skills. In a survival situation you must force yourself to survive; that effort determines the difference between life and death.

✓ DO

- Put on your lifejacket as soon as possible — in cold water you will lose the use of your fingers
- Board the lifeboat or life raft before entering the water when abandoning ship
- If you must enter the water, enter it gradually, in stages
- Adopt the Heat Escape Lessening Position (HELP) if immersion in cold water is unavoidable: knees drawn up to the chest, arms holding the legs
- Try to remain on your back with your feet up, elbows at your sides and arms across your chest, keeping neck and head above the water if possible
- Where several personnel are in the water, form a circle so the water in the middle warms slightly; place hypothermic casualties in the centre

⊘ DON'T

- Do not swim — swimming increases the body's heat loss. Only swim to a safe refuge that is nearby
- Do not discard gloves or mittens, in the water or in a lifeboat
- Do not delay putting on layers of clothing until after the general alarm has been acknowledged
- Do not lift a recovered casualty vertically — recovery from the water should be made in a horizontal manner so the loss of hydrostatic pressure on the heart is less acute
- Do not underestimate morale: survival chances increase when casualties are together, warming and forming a larger target for rescue

Preventative measures against a general cooling of the body

Hypothermia is caused by a general cooling of the body. If body temperature drops much below its normal level of around 36.8 °C, hypothermia will begin to set in. Severe hypothermia can lead to death.

- It is better to wear several layers of loose clothing than one thick piece — the body requires blood to be constantly circulating, particularly to the extremities, and tight clothing can restrict this
- The face, especially the ears, and the hands and feet should be protected
- Boots should be waterproof and insulated
- Wearing a hat or a hard hat liner will keep the whole body warmer, as it reduces the amount of body heat that escapes from the head
- Limit the amount of time outside on extremely cold days; frequent breaks in warm areas are necessary, with enough time allowed for the body to re-warm
- Thermos flasks containing hot drinks can be used while working
- Avoid touching cold metal surfaces with bare skin, as flesh can stick to the metal and freeze



Protect the head

Protecting the head from increased heat loss is a very important part of avoiding hypothermia. In cold conditions personnel can become hypothermic in minutes rather than hours, and self-help capability may be limited because of freezing muscles and limbs.

Frostnip, frostbite and snow blindness



Frostnip

Usually reversible

- Freezing of the top layers of skin tissue, mostly affecting the cheeks, earlobes, fingers and toes
- Symptoms: numbness; top layer of skin feeling hard and rubbery but deeper tissue soft; skin becomes white and waxy
- Treatment: gradual warming — gently blow warm air onto the area, or place the part against another warm body part such as under the arms
- Care must be taken not to rub the affected skin, as this can damage the tissue



Frostbite

Tissue destruction

- The body constricts the blood vessels at its extremities to reduce heat loss, depriving fingers, toes and facial parts of warming circulation
- Ice crystals form inside the skin and begin to destroy the tissue
- Symptoms: skin that is white with a wooden feel all the way through, numbness, possible anaesthesia
- Treatment: restore warmth to the body immediately



Snow blindness

UV keratitis

- Caused when the eyes are left unprotected from ultraviolet rays in bright sunlight reflected from a snow or ice surface
- Causes a painful keratitis — inflammation and swelling of the cornea; similar to sunburn of the cornea and conjunctiva
- May not be noticed until several hours after exposure
- Symptoms: bloodshot and teary eyes, pain, a gritty feeling, swelling, eyes shut
- Only in very severe cases does it cause permanent vision loss

Re-warming a frostbite casualty



Get the casualty warm

While being in a warm area is essential, it is also necessary to put the patient into warm water.



Maximum 43.3 °C for 30 minutes

Use warm water to a maximum of 43.3 °C (110 °F) for a period of about 30 minutes. The water temperature must be monitored and warm water added as the temperature decreases.



Stop when the skin flushes

When the skin becomes flushed, remove the patient from the water — further warming could lead to additional damage.



Dress the damaged parts

Damaged parts should be wrapped in sterile gauze, making sure that fingers and toes are separated if they are the parts affected.



Escalate after 30 minutes

If after approximately 30 minutes feeling has not been restored to the affected parts, further medical attention will be required.



Snow blindness

Where over-exposure has occurred, rest the affected individual in reduced light conditions to allow the irritation to subside. Before exposure, or at the first sign of symptoms, the use of dark glasses or goggles can greatly reduce the severity of the condition.

Wind chill hazards and risk of frostbite

Wind chill is the apparent temperature felt on exposed skin, and is a function of air temperature and wind speed. Cooling is most rapid at the start of any exposure, when the skin is still warm.

Wind chill	Risk of frostbite	Health concern	What to do
0 to -9	Low	Slight increase in discomfort	Dress warmly
-10 to -27	Low	Uncomfortable. Risk of hypothermia if outside for long periods without adequate protection	Dress in layers of warm clothing with a wind-resistant outer layer. Wear a hat, mittens and scarf. Keep active
-28 to -39	Increasing risk: exposed skin can freeze in 10 to 30 minutes	Check face and extremities (fingers, toes, ears and nose) for numbness or whiteness. Risk of hypothermia if outside for long periods without adequate protection	As above, and cover exposed skin: wear a hat, mittens and a scarf, neck tube or face mask. Keep active

In sustained winds over 50 km/h, frostbite can occur faster than indicated. Source: TP 14335 Winter Navigation on the River and Gulf of St Lawrence, Transport Canada.

High risk, warning and danger levels

Wind chill	Risk of frostbite	Health concern	What to do
-40 to -47	High risk: exposed skin can freeze in 5 to 10 minutes	Check face and extremities for numbness or whiteness (frostbite). Risk of hypothermia if outside for long periods without adequate protection	Dress in layers of warm clothing with a wind-resistant outer layer, and cover all exposed skin: hat, mittens and a scarf, neck tube or face mask. Keep active
Warning level: -48 to -54	High risk: exposed skin can freeze in 2 to 5 minutes	Check face and extremities frequently for numbness or whiteness (frostbite). Serious risk of hypothermia if outside for long periods	Be careful. Dress very warmly in layers of clothing with a wind-resistant outer layer. Cover all exposed skin: hat, mittens and a scarf, neck tube or face mask. Be ready to cut short or cancel outdoor activities. Keep active
Danger: -55 and colder	High risk: exposed skin can freeze in less than 2 minutes	Outdoor conditions are hazardous	Stay indoors

In areas with a milder climate, a wind chill warning is issued at about -35 °C. Further north, people have grown more accustomed to the cold: residents of the Arctic are warned at about -50 °C, and those of the high Arctic at about -55 °C.



SECTION 15

Cargo and Port Requirements

Failure to take freezing weather precautions before arrival may prevent the vessel from loading or discharging at all.

15

Why cold-weather precautions must be taken before arrival

Many tanker destination ports are subject to occasional onsets of extremely cold weather and even more extreme wind chill factors, which affect cargo operations. All fleet vessels arriving at such ports must take all necessary freezing weather precautions before arrival to ensure equipment operability and prevent damage.



Fire and foam systems

Cold temperature damage or impairment to fire and foam systems is one of the first consequences of arriving unprepared.



Steam systems

Steam systems may be damaged or impaired, removing the means of heating cargo and tracing lines.



IGS water seals and P/V breaker

The inert gas system water seal and the pressure/vacuum breaker are directly exposed to freezing.



Hydraulic systems

Hydraulic systems for cargo valves, cranes and deck machinery are impaired by cold and by moisture in the oil.



Domestic water supplies

Domestic water supplies may be lost, to the degree that the vessel may be prevented from loading or discharging its cargo.

Temperatures, trims and frequencies at cold ports

Condition	Requirement
Sea water temperature below 0 °C	All water tanks adjacent to the vessel's shell plating should be reduced to 90% capacity to allow for expansion if freezing takes place.
Trim	Restrict the trim of the ship to between one and three metres, to avoid the chance of ice sliding along the bottom of the ship and blocking the engine room intakes.
Freezing spray on passage	When the vessel is experiencing freezing spray during the voyage, tank pressures must be monitored on a 4-hourly basis as a minimum.
Unheated vent heads	All vent heads which are not heated should be checked every 20 minutes.
P/V valves below -15 °C	At temperatures below -15 °C, manual openings of the P/V valves will be applied at 20-minute intervals.
Hose handling crane	The hose handling crane hydraulic motor should run without any load for 1 to 2 hours prior to use.

Keeping the engine room intakes immersed

- If the vessel has fresh water or brackish water ballast, it should be changed with sea water prior to arrival in the cold weather zone
- Vessels proceeding to Canada or the US east coast in winter should change ballast before crossing the western wall of the Gulf Stream
- At the cargo berth, take on ballast as early as possible, berth depth permitting, to keep the engine room intakes immersed as deeply as possible
- Bear in mind that this keeps the wing tanks immersed longer, with increased cargo cooling compared with a lighter draft
- Load cargo prior to discharging ballast to maintain maximum draft
- While ballasting and de-ballasting tanks, keep the sampling hatches open to ensure ventilation is adequate — vents should not be relied upon
- Keep the cargo pump cofferdams well purged and free of moisture



Deck seal heating

When vessels are trading in cold weather areas it is essential that deck seal heating systems are in operation.

Steel, impact resistance and post-ice inspection



Impact resistance is reduced

At extremely low temperatures the impact resistance of steel is reduced. The incidence of structural failures after transiting heavy ice, or making hard contacts with wharfs or fenders, may increase.



Close examination is mandatory

Close examination of any possibly affected area is mandatory.



Inspect immediately on clearing heavy ice

Immediately on clearing heavy ice areas a vessel inspection should be conducted covering cargo compartments and other spaces such as fore peaks, fore deeps, cofferdams and pump rooms, in order to verify hull integrity.



Cumulative damage

Ice damage on ships' hulls is often cumulative, with dents developing over several winters, and plating thickness worn by several millimetres through corrosion and abrasion.

Precautions when loading or discharging in cold weather

Where the weather temperature is equal to or below the melting point of the cargo, cargo may freeze in tanks and lines. Freezing can start in the lines very quickly through the effect of wind, even where the melting point is above the weather temperature.

✓ DO

- Start cargo heating immediately
- Keep live steam on deck; steam valves should be slightly open to avoid freezing
- Consider freezing of drains, gauge intakes and manifold sampling points, and use steam before use
- Cover valve handles with canvas when not in use, to avoid ice build-up
- If the operation is suspended, blow the lines and keep the valves open to ensure cargo in the lines is fully drained to tank
- Keep the cargo hydraulic system jockey pump (pilot pump) always running and ensure hydraulic oil temperature is not lowered to a critical level, where applicable

⊘ DON'T

- Do not rely on the melting point alone — wind can start freezing in the lines even above it
- Do not leave a suspended operation with cargo standing in the lines
- Do not close valve handles into an ice covering — keep them covered and workable
- Do not start unloading conveyor belts or crane wires cold: run all conveyor belts at no load for a warm-up period, and run crane wires periodically at no load to prevent ice build-up

P/V valves, mast risers and IG lines

Referring to ISGOTT cold weather precautions, the vents are checked regularly to maintain the valve operational at all times.



Check lifting cycle

P/V valves must be checked regularly and kept free of snow and ice accumulation. Precautions may include a check lifting cycle that loosens the ice coverings and prevents the valve being blocked by ice.



Twenty-minute intervals below -15 °C

The frequency of manual opening is established to ensure continued operation of the valve and prevent restricted movement through ice formation. At temperatures below -15 °C, 20-minute intervals of manual opening apply.



Canvas covers

P/V valve vent heads may be covered with canvas during the voyage to avoid build-up of freezing spray. For the P/V valves on board fleet vessels, heating arrangements are not required.



Mast vent riser

The mast vent riser valve is to be protected with grease and a canvas cover. Flame arresters are to be free of ice before the start of cargo operations.



IG lines drained and checked

It should be checked prior to arrival that mast risers and inert gas lines are drained of any liquid. Essential valves on IG and venting systems should be protected with suitable grease and a canvas cover, and the operation of piston breather valves on IG lines checked before operations commence — covers removed where applicable.

Portable pumping arrangements in the cold



Wilden pump and MARPOL lines

The fixed Wilden pump and line attached to the port and starboard drip trays and MARPOL lines freezes up very easily, so the vessel must have a portable Wilden pump standing by with sufficient rigid hose to reach the residual tank. Secure the pump and hose where they will not freeze when not in use, and blow the hoses through with air after use.



A second pump aft

Have another portable pump ready on the aft main deck for any cargo spills. The fixed stripping system might become inoperable due to the pump and lines freezing. This pump and hose should also be secured where they will not freeze, and the hoses blown through with air after use.



Gauging and manifold points

Means to prevent ice accumulation on the tank gauging and dipping point valves, and on cargo manifold pressure gauge connections, should be available.



Manifold drip trays

The cargo manifold drip trays should be maintained dry, and the drain valves on the drip tray drained of any water to prevent them freezing in sub-zero temperatures. Exposed valve actuators should be covered with canvas to protect from freezing spray and ice. The portable emergency hydraulic pump for cargo valves should be topped up with oil and tested, and all officers should be familiar with its use.

Deck watch arrangements in port

–25 °C

Below this temperature the Master shall appoint an extra deck watchkeeper to shift the other two watchkeepers

15 min

Minimum rest in the accommodation after each hour of watch outside the accommodation



Additional watchkeeper — An additional watchkeeper may be appointed from the engine department, and may be assigned as a gangway watcher only.



Purpose of the rotation — The rotation protects watchkeepers from the extreme cold and its adverse effects on the body.



Master maintains the arrangement — The Master maintains these items during watch on deck in port.



SECTION 16

Towing Procedures

Any ship requesting icebreaker assistance should expect to be involved in towing, and should have towing gear rigged continually.

16

THE SINGLE BIGGEST CAUSE

Excessive speed is a common factor in the cause of damage to ships by ice

Officers must be aware that the ship's speed should be very carefully selected and impacts with ice avoided. Navigators should know the speed at which damage might occur in the ice conditions being experienced, during towing and during navigation in ice.

Select speed deliberately

Avoid impacts

Know your damage threshold

Requirements for ice class ships designed to tow



Line throwing apparatus

All ice class ships designed to perform dedicated towing operations, and all icebreakers, should be equipped with line throwing apparatus in addition to that required for life-saving, capable of delivering messenger lines for the transfer of towing equipment.



Not powder or rocket type

Such line throwing apparatus should not be of the powder or rocket type, so that it may be safely used to make a transfer to a tanker.



Quick release system

All ice class ships designed to perform dedicated towing operations should be provided with a quick release system, operable from the conning position.



Close-coupled arrangements

Where fitted, close-coupled bow-to-stern towing arrangements should comprise strengthened bow plating on the towed ship, appropriate towing slings, non-interfering positioning of bower anchors, and disallowance of bulbous bows.



Securing the anchor

In that case, arrangements should be provided for securing the anchor in the stowed position.

Receiving assistance

All ice class ships should be capable of receiving emergency towing assistance.



Connection and release

Where appropriate, towing arrangements should facilitate connection and release of a towline, and provide bollards, fairleads and other components suitable for the size of ship on which they are fitted.



Ship-specific details

For specific vessel details, see the Emergency Towing Manual of the ship. Large tankers are fitted with emergency towing brackets, which make the operation much more practical where icebreaker assistance for towing becomes a requirement.



Prepared before the ice edge

Examine the towing fittings on the forecastle and ensure that a suitable messenger is available to pick up a bridle or towline. Anchors are hung off or led away from the hawse pipe so a tow sling can be threaded if required.

Arranging a tow in ice

1

Make the request

A request to be towed may be directed through agents, shipowners or by the Master, and may be made to a Port Authority, the Port Director or Harbour Master, or to the officer commanding the icebreaker itself.

2

Settle the commercial terms first

To avoid fees for salvage, any fees or costs involved should be ascertained before the towing operation is commenced. A written agreement between the interested parties is preferred wherever possible.

3

Agree the towing arrangement

Consider a bow connection or an astern connection, the horsepower of the assisting ship, and the manoeuvrability of the assisting ship. Clear and precise securing details should be clarified at the beginning, to save time and to provide a safe environment for operators.

4

Inform the icebreaker

Notify the icebreaker of areas capable of accepting the towline — a usable strong point, chafing chain or stopper arrangement — and provide details of fairleads, insurance wire, diameter of bitts and the size of any towing bracket fitted.

5

Assess the risk

The Master of the assisted ship should make an overall assessment of the situation and evaluate the risk throughout the proposed towing operation, comparing it to the dangers of remaining beset in ice.

Length, layers and speed

700–1,000 m

Typical towline length in temperate latitudes — not advisable for icebreakers

4 or 5

Maximum layers on the winch drum, to prevent the outer layer being depressed into the lower windings



Why tows in ice are short — Long tows are not possible in ice infested waters, as the channel formed by the icebreaker would quickly clog and increase the resistance of the towed vessel. Tows in ice are much shorter and possibly close coupled as a composite unit.



Speed is the icebreaker's decision — The speed of any towing operation will be at the discretion and order of the icebreaking ship. The assisted ship should make every effort to keep the towline taut despite variations in the icebreaker's speed caused by the ice.



Adjustable length — Ideally, the length of the towline should be adjustable. The leads for the line should be well parcelled and protected against chafing.

Considerations before and during the tow

- Establish the type of towline, with regard to the dangers associated with hawsers close to propellers, particularly when towing from a stern position
- It is preferable to secure the tow during the hours of daylight, but in polar regions this may not be possible
- Consider prevailing weather, communications between the two ships and the navigation signals
- Maintain detailed records of the operation throughout, including times and positions, with appropriate entries made in the ship's logbook
- Make a report to the owners and agents of the assisted ship
- Consider the proposed speed of the towing operation, as this directly affects progress and ETA for the next port
- The route while under tow must be a safe passage with adequate underkeel clearance throughout, charted as a passage plan under tow, on a route that presents no additional hazards
- Keep towing gear rigged continually and ready for immediate use — virtually all icebreakers are fitted with towing winches



Ship-to-ship towing

It is not recommended that commercial ships tow one another unless specially fitted for close coupled towing.

Close towing in compact, heavy or hummocked ice



The ship is pulled close astern

When icebreaker assistance takes place in compact, heavy or hummocked ice, close towing is carried out and the icebreaker pulls the assisted ship close to its stern.



No clear channel exists

In thick ice an icebreaker does not open a clear channel. The ice will close only a couple of metres behind the icebreaker or the towed vessel.



Brash ice forced downwards

The icebreaker's path creates a jumbled brash of ice, and pressure from both sides of the pack closes this in immediately, forcing the brash ice down further than the original caked ice.



Steering is impaired

The icebreaker's steering is impaired because of the weight of the ship following, so course alterations should be carried out in loose ice.



Hang off the anchors

Anchors are hung off or led away from the hawse pipe, to facilitate threading of the icebreaker's tow sling through the anchor hawse pipes should close coupled towing become necessary.



Use a towing bridle

While towing in heavy ice it is advantageous to use a towing bridle that spreads the load onto more than one set of bitts.

Towing into the notch

In thick ice, when the icebreaker has gone ahead, there may be many large pieces of broken ice in the channel. In such circumstances, particularly in the Baltic, the icebreaker will tow the ship by pulling it into a small docking V at the stern — a notch.



How it is rigged

The icebreaker hands over two or four bridles, which are fastened to the towing bollards. The ship being towed has its bow flush against the stern of the icebreaker.



Better chance of passing the ice

The towed ship is directly behind the icebreaker and so has a far better chance of passing through any ice.



Extra mass helps the icebreaker

The icebreaker can benefit from the extra mass the towed ship gives when clearing particularly difficult ice areas.



Collision risk removed

An additional benefit is that the risk of collision has been removed. It should always be considered that close coupled towing requires considerable skills and training.

Operating rules for towing in ice

✓ DO

- Expect to be involved in towing whenever icebreaker assistance is requested, and keep towing gear rigged continually and ready for immediate use
- Ascertain all fees and costs, and prefer a written agreement, before the towing operation is commenced
- Keep the towline taut despite variations in the icebreaker's speed
- Make course alterations in loose ice when under short lead tow
- Use a towing bridle in heavy ice to spread the load across more than one set of bitts
- Expect two icebreakers to be used for a wide-beam tanker — one breaking the track while the other tows

⊘ DON'T

- Do not tow another commercial ship unless specially fitted for close coupled towing
- Do not use long tows in ice — the channel clogs and the resistance of the towed vessel increases
- Do not allow more than 4 or 5 layers on the winch drum
- Do not lead a hawser close to a propeller, particularly when towing from a stern position
- Do not proceed under tow without a charted passage plan and adequate underkeel clearance



SECTION 17

Hull Integrity and Damage Control

In ice-covered waters ships need to be more self-reliant. It can take a long time to reach a safe port, or for help to arrive.

17

The requirement for damage control capability

The IMO Guidelines for ships operating in Arctic ice-covered waters state that all ships navigating in Arctic ice-covered waters should be adequately equipped and their crews properly trained to provide effective damage control and minor hull repair.



Sufficient equipment

Damage control equipment should be sufficient to enable a ship, as far as practicable, to make temporary repairs to a minor hull breach.



Prevent escalation

It should also enable precautionary measures to be taken to prevent escalation of damage or flooding, so that the ship may proceed to a location where more substantial repairs can be effected.



Masters must train the crew

For ice class fleet vessels, damage control equipment is listed in the manuals, and all Masters are responsible for giving the necessary training to crew so that they are well familiar with the requirements and the equipment.



Take the impact on the bow

If there is an unavoidable collision with ice floes, the vessel should be positioned so that the impact will be received by the strongest part of the vessel at the bow.

The first five actions after impact in ice

Although every incident will have its own idiosyncrasies, the following actions should be undertaken following a collision or impact in ice conditions.

1

Stop engines

Stop engines immediately to prevent further contact and to protect the propeller and rudder.

2

Raise the alarm

Raise the alarm and inform the ship's Master, who is expected to take the con.

3

Close watertight and fire doors

Close watertight doors and automatic fire doors.

4

Damage assessment by the Chief Officer

The Chief Officer must carry out a damage assessment covering watertight integrity of the hull, engine room condition (wet or dry), a casualty report and pollution.

5

Chart the position

The ship's position should be charted. After the initial damage assessment, an urgency or distress message may need to be issued by radio broadcast.

Actions once the assessment is complete

- If the hull has been breached and water ingress is taking place, order the pumps onto the affected area
- If the damage is on or near the waterline, transfer ballast to generate an adverse list, to bring the hull damage to a position above the waterline
- Order survival craft to be turned out and made ready for launch if the damage is severe enough to cause sinking
- Gather together and employ damage control parties
- Carry out a detailed inspection, to include tank soundings of the affected area and adjacent spaces
- Once the tank disposition is known, make a stability assessment of the ship's survivability
- Investigate the possible use of a collision patch on any damaged area
- Display appropriate navigation signals, for example not under command, and alert coast guards
- Provide medical aid to any casualties as soon as is practical
- Assess the navigation situation with regard to the nearest port of refuge, or shallows for beaching, to prevent the ship's loss



Preparing for abandonment

- Prepare life craft with extra blankets, clothing and food
- Immersion suits ready
- Rescue boat and lifeboat engines turned over and warmed through prior to launch
- Note that launching devices may have to be de-iced prior to use

Reports to be made when time allows

Detailed records of the incident, including times and positions, support the investigation, the insurance position and the decision on where substantial repairs will be made.



Owners

Provide relevant details of the incident, together with the support and assistance required.



MAIB

Report to the Marine Accident Investigation Branch.



MPCU

Report to the Marine Pollution Control Unit where pollution has occurred.



The agents

To inform charterers and arrange repair facilities if required, and to revise ETA and future progress.



Communications

Update priority communications to include downgrading or cancelling distress.



Damage record

Record damage details to the ship and to any cargo.



Logbook entry

Make an entry into the ship's logbook.



Emergency services

Coordination with emergency response services should be established in accordance with M4 Emergency Manual requirements.

COUNTER-INTUITIVE BUT CRITICAL

The smaller the hole in the hull, the stronger the force of water entering the ship

While catastrophic damage can still result, ice damage below the waterline can be small and still capable of allowing water to jet into the ship. While the pumps could probably cope with the flow, any ingress is undesirable — and if the penetration is in the engine room, damage to machinery and electrics can render the essential pumps inoperative.

Small hole, high pressure

Protect the machinery

Fix it temporarily, fast

Materials for temporary repair of holes and leaks

A selection of materials for the temporary repair of holes and leaks should be available to the crew.



Timber

For shores, wedges and backing pieces.



Canvas and patching materials

For covering a breach from outside or inside.



Bolts and washers

To secure patches over a damaged area.



Oakum

A loose fibre obtained by unravelling old rope, used for filling cracks.



Quick drying cement

To set a box or patch over a small breach.



Screw jacks and stanchions

Screw jacks and portable stanchions to hold shoring in place.



Steel wire and cordage

Steel wire and cordage ropes for securing and hauling.



Portable welding equipment

For permanent-style repair where conditions allow.

Three temporary damage control devices that still work

There are a number of temporary damage control devices that may seem to have come from a bygone age, but that could still prove very effective.



Wooden wedges

Both flat and pointed. These are driven into the hole, singly or in multiples, to block it and limit the water entering. A box kept in the engine room and other strategic places during ice transit would be ready for instant use.



Weighted canvas or tarpaulin

A large, bottom-weighted canvas or tarpaulin sheet with ropes attached, stowed in a locker on deck ready to be thrown over a damaged area. The weighted bottom drops the sheet down alongside the hull and the force of the water drags it towards the damage, forming a cover over the hole.



Why the sheet helps

This does not completely prevent water entering, but the force of ingress is greatly reduced, enabling those inside to take more effective action. It is difficult to place wedges against the force of the water, so any let-up in the flow is welcome.



Survival suits in the engine room

A number of survival suits provided in the engine room, as part of the damage control equipment, will greatly ease the discomfort for crew members working under a jet of icy water. Easy access could save critical minutes.



A splinter box

A dome shaped device with rubber seals around the rim, and a long screw through the dome with a folding bar. They come in various sizes and are present in the damage control inventory on warships. A splinter box is a heavy metal box used to patch a hole made in the side of a ship.



SECTION 18

Causes of Damage in Ice

Any ship engaged in passage through ice infested waters must expect a degree of damage. Mariners should know the areas of vulnerability.

18

What damage in ice means for the ship and the voyage



A port can be unsafe because of ice

A port can be considered unsafe by reason of the ice present on the route to that port, and whether or not it can be kept open by the presence of icebreakers.



Expect a degree of damage

Any ship engaged in passage through ice infested waters must expect a degree of damage, caused either by contact with ice or by the extreme cold temperatures being experienced.



Protection of the hull is paramount

Any large piece of ice, for example an ice cake or a growler, striking the shell plate could breach the watertight integrity of the ship. The position of any ice at the waterline is a continuous threat.



Even thin ice wears the coatings

Even a passage through comparatively thin ice will wear the protective paint coatings of the hull in both the upper and lower regions of the waterline.

Channel hazards and the two factors that matter



Hazards in a new channel

- The edges of channels cut by an icebreaker are often left with projecting rams of ice
- There is a high possibility of the assisted ship's hull colliding with these projections, resulting in puncture damage to the outer shell
- Newly opened channels may also contain isolated floes left behind astern of the icebreaker
- These floes may obstruct the forward progress of following ships



What watch officers must watch

- The speed of the ship
- The thickness of the ice through which the vessel is navigating
- Early avoidance of ice concentrations is by far the better line of action
- Such tactics may not always be possible or practical

Temperature, speed and collision



Below freezing temperatures

At low temperatures the impact resistance of steel is reduced. Structural damage can occur at a greatly reduced level of impact with ice, docks or fenders.



High speed

This applies particularly to ships of high propulsion power relative to their size, which may have open water speeds of over 20 knots. Striking an ice edge at full speed may damage the bow area. Moving fast in an ice channel may damage the side shell and plating above the waterline on the bow, through extended ice loading by the bow wave. On such ships, slow manoeuvring speeds may still be faster than what is considered a safe speed in ice.



Collision with another ship in a channel

When ships move in an ice channel one after the other with a short lead, if the first ship becomes stuck in ice the next one may collide with it. This is more likely if the first ship is small with low propulsion power and the following ship is much bigger.



Damage during icebreaker assistance

A collision with an icebreaker can occur with short lead assistance if the icebreaker becomes stuck in ice. Damage may also occur in towing and cutting situations if the icebreaker cuts the ship too close, and if the ice intended to form a spacer between the vessels either collapses or moves away.

Channel contact, reversing, compression and ridges



Damage in an ice channel

Mid and aft ship hull areas may be in contact with the channel edges when the ship is turning. Bilge and bilge keels may be damaged, particularly when the channels are old with thick edges.



Reversing in ice

In heavy ice conditions ships often become caught in ice and have to back out. Reversing in ice may damage aft ship areas, particularly the rudder and propeller. Aft areas of hulls may also be damaged through manoeuvring in ice.



Compressive ice

The ice channel may close due to the moving ice field. In this situation the ship becomes stuck in ice, and damage often occurs in the midship area on the flat side region.



Ice ridges

Ice ridges can extend 3 m above and 20 m below the water level. Fragments can damage side, bilge and bottom areas, particularly when a ship is towed through one. Seawater wells in the bottom of the ship may become plugged by small ice pieces from the ridge. Plugging of sea bay inlets can also occur from prolonged exposure to brash ice conditions, as seen in a river system.

Port manoeuvring, darkness, condition and experience



Manoeuvring in port

Ice rubble in port can be several metres thick and contain objects such as rocks and wood. Manoeuvring in this ice rubble can damage many areas of the ship's hull and parts of the propulsion machinery.



Darkness and snowfall

During the northern winter there is very little daylight available for ice navigation. Combined with snowfall, wind and freezing temperatures, this makes navigation in the Baltic in winter much more difficult than in summer.



Bad repair or insufficient strengthening

Ice damage on hulls is often cumulative, with dents developing over several winters. Ships may have relatively large deformations on plating and frames, and plating thickness may have worn by several millimetres through corrosion and abrasion. Ships without correct ice strengthening or in bad repair are more likely to suffer serious damage.



Crew inexperienced in ice navigation

During the 1990s ice conditions in the Baltic Sea were relatively easy for ice navigation. Experience of winter navigation may have been forgotten, or may never have developed.

Icing, grounding and vibration



Icing effect

Icing of the ship's superstructure is caused by the lifting of sea spray into the wind at the ship's bow. The spray is super-cooled and carried over the superstructure to freeze on bulkheads, decks and rigging. This is a particular problem for small ships such as fishing vessels and tugs.



Grounding events

A grounding event may occur due to the drifting of a ship within a moving ice field. It can also occur when a ship seeks an easier route in difficult ice conditions and then accidentally grounds.



Vibration

Vibration of the structures due to constant manoeuvring at high revolutions, and the possibility of milling ice through the propellers.

Damage above the deck

Upper structures that experience ice accretion may also suffer a degree of damage.



Aerials and halyards

Aerials, halyards and similar upper rigging can become overburdened with the added weight of ice, and the aerials may give way under heavy load.



Unlagged water pipes

Water pipes that remain unlagged may freeze and crack.



Fresh water tanks in survival craft

Fresh water tanks, particularly any in survival craft, can exhibit damage from ice expansion when exposed to freezing conditions.



How to avoid it

Cracking or damage of water tanks can often be avoided by reducing the water content to allow for expansion, or by keeping an alternative supply in a warm space.



SECTION 19

Oil Pollution in Ice-Covered Waters

An accidental spill represents one of the most serious potential threats to ecosystems that could occur as a result of shipping.

19

What the ship's crew does first in any response scenario

The priority for the crew in any response scenario will always be to ensure the safety of the vessel first and foremost, and to stop or reduce any oil release as much as possible.



Damage and stability control

A combination of damage and stability control to keep the ship intact and upright.



Move to a safe haven

Moving to a safe haven, if one is available.



Temporary repairs

Temporary repairs to minimise oil loss and water ingress.



Internal transfer

Transferring fuel or oil cargo to undamaged tanks.



Coordination ashore

Coordination with emergency response services should be established in accordance with M4 Emergency Manual requirements.

The behaviour and fate of oil in a cold environment



Immediate and long-term consequences

An accidental spill is one of the most serious potential threats to ecosystems that could occur as a result of shipping. The release of oil could have immediate and long-term consequences.



Oils differ greatly

Crude and refined oils, including fuel oils used by ships, vary greatly in their physical and chemical properties. These properties, together with temperature, waves and ice, largely determine the extent of long-term biological effects.



No infrastructure

Responding to oil spills in such areas, under often extreme climate conditions, will always be a major challenge, exacerbated by the lack of any basic infrastructure.

The key response strategies for oil in ice

Spill response operations in ice and open water are fundamentally different. Over-reliance on any single response strategy is bound to fail, given the infinite variety in ice environments and oil-in-ice situations that may be encountered in a real spill.



Mechanical recovery

Well understood

Recovery with booms and skimmers, and recovery without booms for small localised spills. Operating windows, constraints and effectiveness are well understood.



In-situ burning

Well understood

Burning with and without fire booms, as an offshore response. Also well understood in terms of operating windows, constraints and effectiveness.



Chemical dispersants

Relatively new

Applied from the air or with vessel spray booms. Relatively new and still uncertain in terms of applicability. Vessels cannot use any dispersants without permission.

How incidents and accidents are grouped

The incidents and accidents recorded in ice-covered waters were grouped into the following five categories.



Grounding

Contact with the seabed, often through drift in a moving ice field.



Collision

Contact with another ship, an icebreaker or a heavy floe.



Fire or explosion

Heightened by extra heaters and electrical load in the cold.



Sunk or submerged

Total loss, with oil released from the wreck.



Machinery damage or failure

Including loss of propulsion from blocked sea inlets.

How vessel spill scenarios are categorised

The combination of these elements, together with the ice conditions, controls the behaviour of the spill in the hours and days following the incident, and determines the response strategy with the best chance of containing, removing or recovering the oil.



Oil type

- Diesel
- Heavy fuel oil
- Refined products
- Crude oil



Spill source

- Sudden batch release
- Chronic leak

What happens when oil meets snow

Oil on ice and in freezing leads will adsorb large volumes of snow over time. This process creates a mixture that is surprisingly dry.

60–80%

Snow content at which additional snow covers the oil and hides it from sight



It can still be burned — This oil and snow mixture can be effectively burned on a solid ice surface, but only if crews and equipment can be safely deployed onto the ice to concentrate the oiled snow.



Nearshore only — This strategy is most applicable near shore, where there is a high assurance that the ice sheet will remain stable.



Natural containment — The most dramatic difference between spills in ice and open water is the spreading behaviour. In many situations, oil spilled into ice will be naturally contained within a relatively small area — which has mostly positive implications for response times and recovery options.

Five scenarios leading to accidental spills in ice



Small Tier 1 spills on deck

Resulting from burst pipes or drums on the decks of a ship. This oil is normally contained on board, or results in minimal over-the-side discharge.



Leakage into ballast systems

Oil leaking into ballast systems which are subsequently discharged overboard, leading to potentially large fines through airborne surveillance.



Transfer spills at terminals

Spills during oil loading or unloading at terminals due to a breaking hose or an open valve. These would be moderate, from a few litres to a few cubic metres, as pumping would be stopped once leakage was noticed. Although relatively small, transfer spills are the most common form of accidental spill.



Tanker accident — Tier 3

A tanker accident along a main shipping route could cause the most severe pollution, for example where multiple compartments are penetrated through a high-energy collision with an iceberg or growler, or through grounding. Although highly unlikely, in the worst case such an internationally significant Tier 3 incident could result in tens of thousands of tonnes of crude oil being spilt.



Loss of bunkers

Loss of bunkers from structural break-up, collision, grounding or explosion.

Natural containment and migration of oil in ice

The following assumptions can be used to gauge the potential effectiveness of different response strategies in different spill scenarios.

Situation	Assumption
Oil between floes	Natural containment occurs at any time of year in concentrations consistently greater than 6/10 — that is 60% area coverage.
At freeze-up	Natural containment is more likely with grease, nilas or new ice at freeze-up.
Oil on or under the ice	Natural containment also occurs for oil deposited on top of the ice, and for oil deposited under the ice.
Spring warming	Natural migration of oil through the ice is likely when the ice warms in the spring, April to May, depending on latitude.
Thin ice at freeze-up	Natural migration is likely when the ice is less than 30 cm during the freeze-up period, depending on brine volume and ambient temperatures.

Mechanical recovery by ice concentration

Scenario	Effectiveness
Large spills into water, with booms	Effective as long as regional ice concentrations are 1/10 or less, that is open water. The principal constraint will be sea state, as in conventional spill response.
Up to 3/10 with booms	Still possible, but much less effective in concentrations up to 3/10.
Over 3/10 with booms	Not effective with regional ice concentrations over 3/10.
Small localised spills, without booms	Effective in concentrations of 7/10 to 9/10, where the oil is naturally contained.
Small spills at 4/10 to 6/10	Possible, but expect a significant portion of the spill to spread into films too thin for recovery without booming.
Large spills on or under fast ice	Effective with surface vehicle access to shore.

Mechanical recovery effectiveness in ice is seriously degraded in any ice cover over 1/10, due to limited encounter rates and ice interference with boom and skimmer systems.

In-situ burning without fire booms

Scenario	Effectiveness
Large spills between floes	Effective with ice concentrations of 7/10 or more; possible in future with herding agents in lower ice concentrations.
Large spills onto the ice surface	Effective with aerial ignition or surface access in fast ice or pack ice, as long as there is sufficient oil film thickness.
Oiled snow	Oiled snow can be concentrated on the ice surface for burning, but this can only be achieved on fast ice nearshore and with relatively thick oil films to begin with.
Oil under ice from a sunken vessel	Effective with new or young ice at freeze-up, or warm ice in spring and summer, when oil surfaces as thick films suitable for aerial ignition. Also effective with stable nearshore fast ice where mechanical equipment can expose the trapped oil.
Offshore trapped pockets	Possible offshore if thick pockets of trapped oil can be located and exposed by helicopter-supported crews with lightweight drilling or cutting equipment.
Slow release under moving ice	Not effective — the resulting oil pockets will be too small and thin to burn or recover.

What works, and what the limits are



Dispersants

The application of chemical dispersants is recognised worldwide as an environmentally acceptable spill response strategy for spills in open water, and a highly efficient means of rapidly eliminating spilled oil offshore under the right conditions. Effective in open water where regulations permit — principally Norway in deep water, and possibly Alaska in future. Not considered operational for spills in ice at this stage. Vessels cannot use any dispersants without permission.



Booms and skimmers

The actual length of boom that can be deployed and maintained under freezing conditions, even briefly, depends on the severity of the ice conditions. Any limited containment possible in very open drift ice requires rugged, high-strength booms to withstand contact. Skimming oil from isolated pools contained by ice is a proven technique for small spills, or as a mopping-up strategy for outlying areas.

What restricts response work in the cold



Safe exposure time

Freezing temperatures offshore will limit the safe exposure time of personnel tasked with operating equipment, monitoring operations and performing operational adjustments and routine maintenance.



Productivity falls sharply

Productivity and effectiveness decline dramatically at cold temperatures, especially with the added effects of wind chill.



Peak gusting winds

Additional weather limitations include peak gusting winds, which will limit crane operations used to load equipment onto barges and lower skimmers into the water.



Nature takes over

As ice concentrations increase and containment with booms becomes increasingly impractical, the natural containment provided by the packing of individual ice floes eventually takes over and limits the oil spread.

What to remember about oil in ice



Mechanical recovery degrades quickly

Mechanical recovery effectiveness in ice is seriously degraded in any ice cover over 1/10, due to limited encounter rates and ice interference with boom and skimmer systems.



Isolated pools can be recovered

Mechanical recovery of isolated pools naturally contained in ice over 6/10 can be effective for small spills, in the order of a few tonnes.



Dispersants need field validation

The use of dispersants in pack ice shows promise in tank testing but needs field validation before being adopted as an operational strategy.



Burning is the most reliable

In-situ burning, with and without herders, is the most effective and reliable means of removing large volumes of oil naturally contained in leads or between floes in pack ice.



The crew's first assessment matters

Although a crew is unlikely to be directly involved in anything but the smallest Tier 1 spills, the crew's assessment of the situation in the initial stages is extremely important — particularly if the vessel is still intact and able to support the recovery effort.



Prepare the ship, the systems and the people

Ice is manageable when the risk assessment is done, the checklists are completed before the ice edge, and every watchkeeper knows the temperature limits that apply to their equipment.



Complete the risk assessment and send it ashore



Know your temperature and freezing limits



Watch the sea inlet pressures continuously



Rotate exposed crew and protect the extremities